

UCS503- Software Engineering Lab

UCS 503 Software Engineering Project Report

DREAMS2REALITY: A minimalist two electrode Software for real-time sleep- stage and dream-state Neural decoding.

Mid-Semester Evaluation

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1. Project Selection Phase

1.1 Software Bid

1. Executive Summary

The **Dreams2Reality** project proposes an innovative, non-invasive software and machine learning system built to decode and identify sleep-stage and dream-state transitions. Unlike legacy clinical polysomnography (PSG) setups that demand 32 to 128 wet-electrode channels, prolonged 45minute preparation, and exorbitant capital expenditure (> \$15,000), our software architecture is optimized for an ultralightweight, two-electrode dry temporal montage (T7–T8). By employing statistical Analysis of Variance (ANOVA F-score) channel selection and Euclidean Covariance Alignment (EA), Dreams2Reality can eliminate inter-subject skull impedance variability without requiring daily recalibration.

2. Problem Statement & Market Opportunity

Over 30% of the global population experiences chronic sleep disturbances, PTSD-induced recurring nightmares, night terrors, and REM sleep behaviour disorders. In clinical psychiatric diagnostics currently, clinicians rely almost exclusively on subjective retrospective self-reports obtained from patients the following morning. These reports suffer from high recall decay, emotional bias, or memory loss.

While clinical polysomnography records (64-128 electrode channel setup) physiological signals overnight, the physical encumbrance of dozens of scalp wires and adhesive paste induces the well-documented **"first-night effect,"** wherein the patient's sleep architecture is severely fragmented by physical discomfort, obscuring natural rapid eye movement (REM) episodes which can cause natural irritation and can affect natural sleep.

These arguments themselves highlight the need to develop software-centric monitoring platform that continuously streams raw EEG signals and pass them dual chamber cross temporal network for gathering info related to different dream-state and sleep stage and to catch their dream issue or traumas.

[Clinical Bottle neck]
64-128 Wet Electrodes, 45-min Setup

[Dreams2Reality Disruption]
2 Dry Temporal Electrodes (T7-T8), <1 min

Alters Sleep Architecture ("First-Night") Natural Undisturbed REM Sleep & Dream Logging

3. Technical Feasibility & Innovation

Our proposed software system demonstrates solid technical feasibility grounded in recent neuroscientific literature and modern TinyML principles:

The GAP in current technology that we tried to cover up→

- 1. Electrode Dimensionality Reduction:** Clinical studies demonstrate that Rapid Eye Movement (REM) dream states and sensorimotor motor hallucinations produce distinct localized power shifts in the bilateral temporal and fronto-temporal

regions. By isolating channels T7 and T8, our pipeline captures temporal beta/gamma desynchronization and asymmetric theta dynamics without requiring whole-scalp coverage.

2. **Cross-Subject Domain Adaptation:** The principal obstacle in non-invasive EEG software is inter-subject variability arising from differences in cranial bone thickness, hair density, and electrode contact impedance. Dreams2Reality addresses this by projecting spatial covariance matrices into a standardized Euclidean reference space: $\tilde{C}_i = R^{(-1/2)} \cdot C_i \cdot R^{(-1/2)}$ where R represents the reference Riemannian geometric mean covariance of the calibration baseline, enabling zero-shot transfer across distinct patients.
1. **Software-Centric Simulation Engine:** For the UCS503 engineering evaluation, physical hardware dependency is abstracted via an asynchronous telemetry playback server. The engine ingests clinical polysomnographic recordings from the PhysioNet Sleep-EDFx benchmark database, streaming 128 Hz packetized data over WebSockets to mimic an active bedside device.

4.Economic and Operational Feasibility

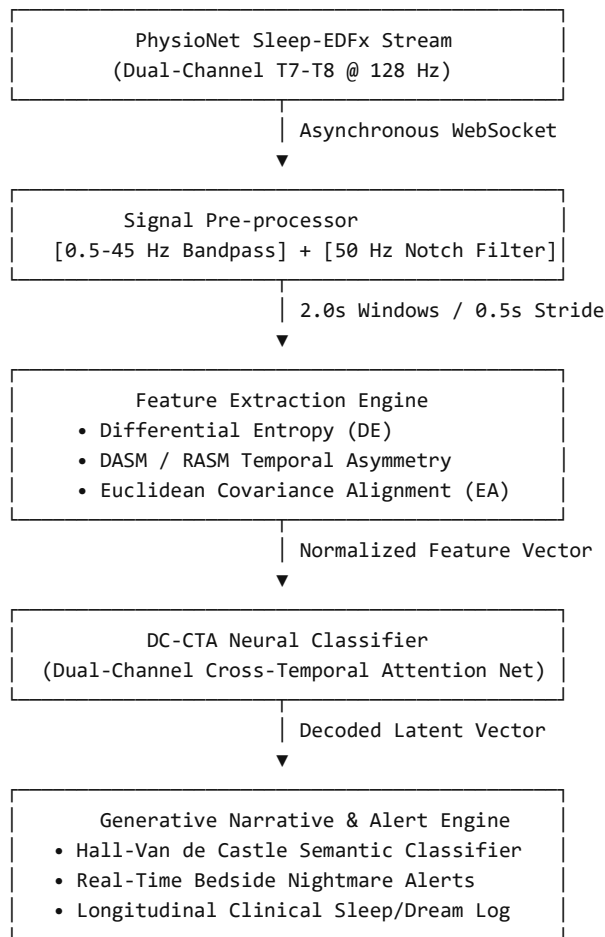
Development Cost: The software utilizes open-source programming stacks (Python, PyTorch, SciPy, MNE, FastAPI, React/WebSockets), resulting in zero software licensing expenditure.

Operational Viability: The inference engine requires fewer than 0.65 MFLOPs per 2.0-second sliding window, allowing execution on modest edge microprocessors (such as Raspberry Pi 4/5 or local bedside PC) with an average latency of < 15 ms, ensuring zero cloud-dependency and strict patient data confidentiality.

1.2 Project Overview

System Concept

Dreams2Reality is an intelligent, real-time biomedical software pipeline that bridges neurophysiological data acquisition and clinical cognitive assessment. The software continuously processes dual-channel EEG data, detects sleep-stage hypnograms (Wake, N1, N2, N3, REM), extracts high-dimensional non-linear features, and decodes the latent emotional valence, arousal, and semantic archetype of ongoing dream states.



Core Architectural Pillars

1. Telemetry controller & Signal Acquisition Subsystem:

Simulates real-time digital EEG telemetry using an asynchronous playback daemon reading calibrated Sleep-EDFx EDF records. It applies 4th-order zero-phase Butterworth filtering (0.5–45 Hz) and adaptive baseline drift suppression.

1. Feature Engineering & Dimensionality Subsystem:

Calculates Differential Entropy (DE) across four standard clinical sub-bands:

- **Theta (4–8 Hz):** Correlated with REM dream vividness and hypnagogic imagery.
- **Alpha (8–12 Hz):** Reflects cortical idling and calm restorative states.
- **Beta (12–30 Hz):** Indicates motor cortical activation and cognitive processing during dreams.
- **Gamma (30–45 Hz):** Captures multi-sensory visual and emotional integration.

Computes Differential Asymmetry (DASM) and Rational Asymmetry (RASM) between temporal channels (T7 and T8) to monitor affective valence.

1. Machine Learning Inference Subsystem:

Employs a lightweight Dual-Channel Cross-Temporal Attention Network (DC-CTA Net) that maps temporal dynamics into five standardized dream mentation archetypes derived from the Hall-Van de Castle clinical coding system.

1. Presentation & Clinical Web Dashboard Subsystem:

A responsive web portal providing real-time hypnogram rendering, raw wave inspection, live trauma/nightmare spike alerts, and an automated Morning Clinical Dream Summary Report.

2. Analysis Phase

2.1 Use Cases

2.1.1 Use-Case Diagrams

The system models interactions between three distinct actors:

1. **Sleeper / Patient (Data Source):** The monitored individual providing simulated or recorded real-time neuro-electrical telemetry.
2. **Clinician / Sleep Therapist:** Medical professional reviewing overnight hypnograms, inspecting detected nightmare episodes, and evaluating patient sleep quality.
3. **System Administrator:** Technical operator managing system calibration parameters, database backups, and simulated stream feeds.

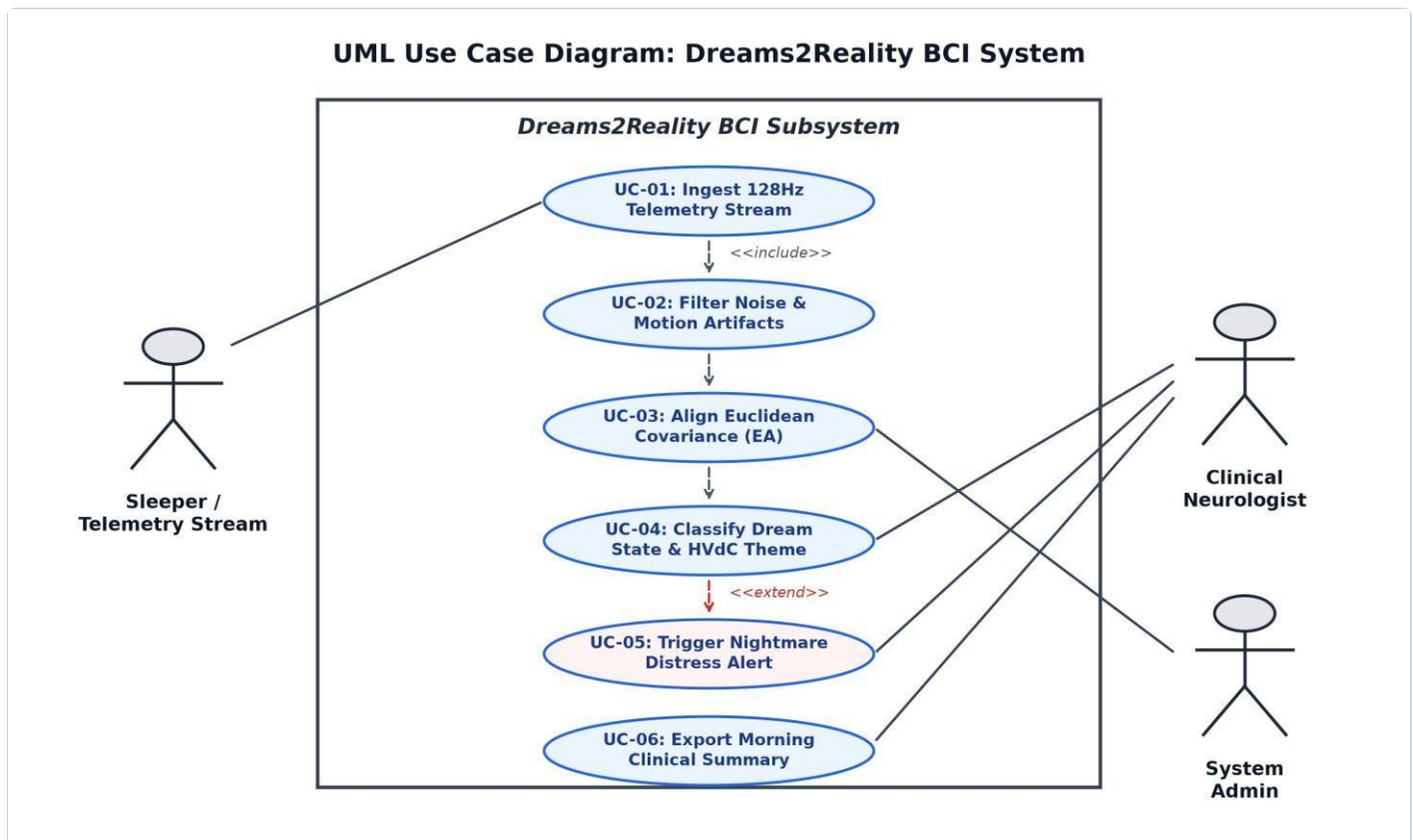


Figure: Dreams2Reality Use Case Diagram — Actor interactions across 8 use cases

2.1.2 Use Case Templates

Use Case Template 1: UC-01 Stream & Ingest Dual-Channel EEG Telemetry

- **Use Case ID:** UC-01
- **Use Case Name:** Stream & Ingest Dual-Channel EEG Telemetry
- **Primary Actor:** Sleeper
- Patient: Desires zero physical intrusion and reliable data capture.
- Clinician: Requires uninterrupted, artifact-free continuous signal logging.
- **Pre-conditions:** The telemetry controller is initialized with a valid PhysioNet Sleep-EDFx record, and WebSocket controller.

Main Success Scenario (Normal Flow):

1. System initializes WebSocket endpoint/ws/eeg-stream.
 2. Telemetry daemon establishes persistent connection and begins streaming packets at 128 samples/second.
 3. Pre-processing buffer acknowledges receipt and verifies packet sequence checksums.
 4. System logs session start timestamp and begins sliding-window assembly (256 samples per 2.0-second epoch).
-

Use Case Template 2: UC-02 Filter Noise

Use Case ID: UC-02

Use Case Name: Filter Noise & Motion Artifacts

Primary Actor: Signal Pre-processing Engine

Secondary Actor: Clinical Neurologist

- **Pre-conditions:** Raw EEG telemetry packets are continuously arriving from UC-01 into the local processing buffer.
- **Post-conditions:** Clean, artifact-free 2.0-second EEG windows are pushed to the feature extraction pipeline.
- **Trigger:** The raw data buffer reaches a 256-sample threshold (representing one complete 2.0-second window at 128 Hz).

Main Success Scenario (Normal Flow):

1. The system receives a 256-sample data block from the buffer queue.
2. The system applies a Butterworth bandpass filter (0.5–45 Hz) to isolate valid biological sleep frequencies.
3. The system applies a 50 Hz notch filter to eliminate localized powerline electrical interference.
4. The system calculates the peak-to-peak amplitude across the filtered 2.0-second frame.
5. The system confirms the amplitude falls within nominal parameters (within $\pm 250 \mu\text{V}$).
6. The system forwards the clean frame downstream to the feature extraction and classification modules.

Use Case Template 3: UC-03 Euclidean Covariance Algorithm

Use Case ID: UC-03

Use Case Name: Align Euclidean Covariance (EA)

Primary Actor: Euclidean Alignment Module

Secondary Actor: System ML Inference Engine

- **Pre-conditions:** UC-02 outputs a clean 2.0-second EEG window, and a baseline reference covariance matrix R is loaded in system memory.
- **Post-conditions:** The patient's spatial brainwave covariance matrix is aligned to a standardized reference space.
- **Trigger:** The pre-processor successfully finishes filtering a data window and emits a pipeline completion signal.

Main Success Scenario (Normal Flow):

1. The module receives the clean 2.0-second EEG signal array.
2. The module computes the spatial covariance matrix (C) across channels T7 and T8.
3. The module applies Euclidean Alignment transformation to normalize cross-subject signal variance:
 $C_{aligned} = R^{(-1/2)} * C * R^{(-1/2)}$.
4. The module verifies matrix symmetry and positive-definiteness post-transformation.
5. The module passes the aligned, normalized covariance matrix directly to the ML Inference Engine for subject-independent classification.

Use Case Template 4: UC-04 Real-Time Dream-State & Semantic Category Classification

- **Use Case ID:** UC-04
- **Use Case Name:** Real-Time Dream-State & Semantic Category Classification
- **Primary Actor:** System ML Inference Engine
- **Secondary Actor:** Clinical Neurologist / Sleep Therapist

Pre-conditions: Ingestion buffer contains at least one calibrated 2.0-second sliding epoch (256 samples) free from catastrophic gross motion saturation ($> 250 \mu V$).

Post-conditions: Classified sleep stage (Wake, N1, N2, N3, REM), affective valence score $[-1.0, +1.0]$, arousal index $[0.0, 1.0]$, and Hall-Van de Castle semantic theme are persisted to the database and broadcast to the dashboard.

Trigger: Pre-processor signals completion of Differential Entropy (DE) and Euclidean Covariance Alignment (EA) computations.

Main Success Scenario (Normal Flow):

1. Inference engine extracts 16-dimensional feature vector (DE across Theta, Alpha, Beta, Gamma bands for T7 and T8, plus DASM/RASM asymmetry).
2. Euclidean aligner projects spatial covariance matrix to reference manifold.

3. Pre-trained DC-CTA Net evaluates tensor input and computes posterior class probabilities.
4. Engine checks for REM sleep criteria and evaluates affective valence.
5. Decoded state is logged to `dream_events`
6. `datastore` and transmitted via WebSocket to the live interface.

Extensions / Alternative Flows:

- 4a. High-amplitude gross motion artifact detected (e.g., patient repositioning):
 1. Threshold filter flags window as corrupted.
 2. Inference engine suppresses classification for the corrupted window.
 3. Dashboard records a "Movement Artifact" epoch without updating dream classification scores.
-

Use Case Template 3: UC-05 Automated Bedside Nightmare & Trauma Distress Alert

- **Use Case ID:** UC-05
- **Use Case Name:** Automated Bedside Nightmare & Trauma Distress Alert
- **Primary Actor:** System Alert Dispatcher
- **Secondary Actor:** Clinical Neurologist / Bedside Attendant
- **Pre-conditions:** Active sleep monitoring session is in progress; patient is classified within REM sleep.
- **Post-conditions:** Visual and acoustic priority alert is generated on the clinician console, logging timestamp, duration, and distress magnitude.

Trigger: Affective valence drops below -0.75 and arousal exceeds 0.85 concurrently with sensorimotor beta/gamma surges for > 10 consecutive seconds.

Main Success Scenario (Normal Flow):

1. System detects sustained negative valence spike coupled with autonomic arousal during active REM sleep.
 2. Distress classifier confirms Hall-Van de Castle category `Threat_Simulation_Panic`.
 3. System triggers priority-1 visual strobe on the bedside dashboard.
 4. System records exact EEG snippet (30 seconds pre- and post-onset) for retrospective clinical evaluation.
 5. Clinician acknowledges alert and applies clinical protocol (e.g., gentle wake-up or calming intervention).
 -
 - **Extensions / Alternative Flows:**
 - 5a. *Clinician does not acknowledge alert within 60 seconds:*
 1. System escalates alarm volume and sends automated push notification to attending nurse terminal.
-

Use Case Template 4: UC-06 Generate Morning Clinical Dream Summary Report

- **Use Case ID:** UC-06
- **Use Case Name:** Generate Morning Clinical Dream Summary Report
- **Primary Actor:** Clinical Neurologist / Sleep Therapist
- **Pre-conditions:** Overnight monitoring session is terminated (minimum 4 hours of recorded telemetry).
- **Post-conditions:** Formatted clinical PDF summary report containing hypnogram timeline, total REM duration, nightmare frequency, and dream semantic distribution is generated and saved.
- **Trigger:** Clinician selects "End Session & Generate Morning Report" on the console.
- **Main Success Scenario (Normal Flow):**
 1. System closes active telemetry stream and flushes remaining buffer.
 2. Analytics engine aggregates all epoch classifications into overnight sleep macrostructure statistics.
 3. Generates graphical hypnogram showing sleep-stage transitions across the night.
 4. Computes total dream duration, percentage of REM sleep, and trauma intrusion index.
 5. Translates numeric event logs into structured natural language paragraphs describing overnight dream trajectory.
 6. Compiles summary into a downloadable, print-ready clinical report.

2.2 Activity Diagram and Swimlane Diagrams

The operational lifecycle of **Dreams2Reality** spans four concurrent execution partitions:

1. **Simulated Telemetry Streamer:** Ingests raw data from calibrated datasets and feeds packets at 128 Hz.
2. **Signal Preprocessing Engine:** Applies digital filtering, artifact gating, and sliding-window buffering.
3. **Machine Learning Brain Engine:** Computes non-linear features, projects Euclidean alignment, and executes neural inference.
4. **Clinical Web Dashboard:** Renders real-time telemetry, hypnogram progression, alerts, and report generation.

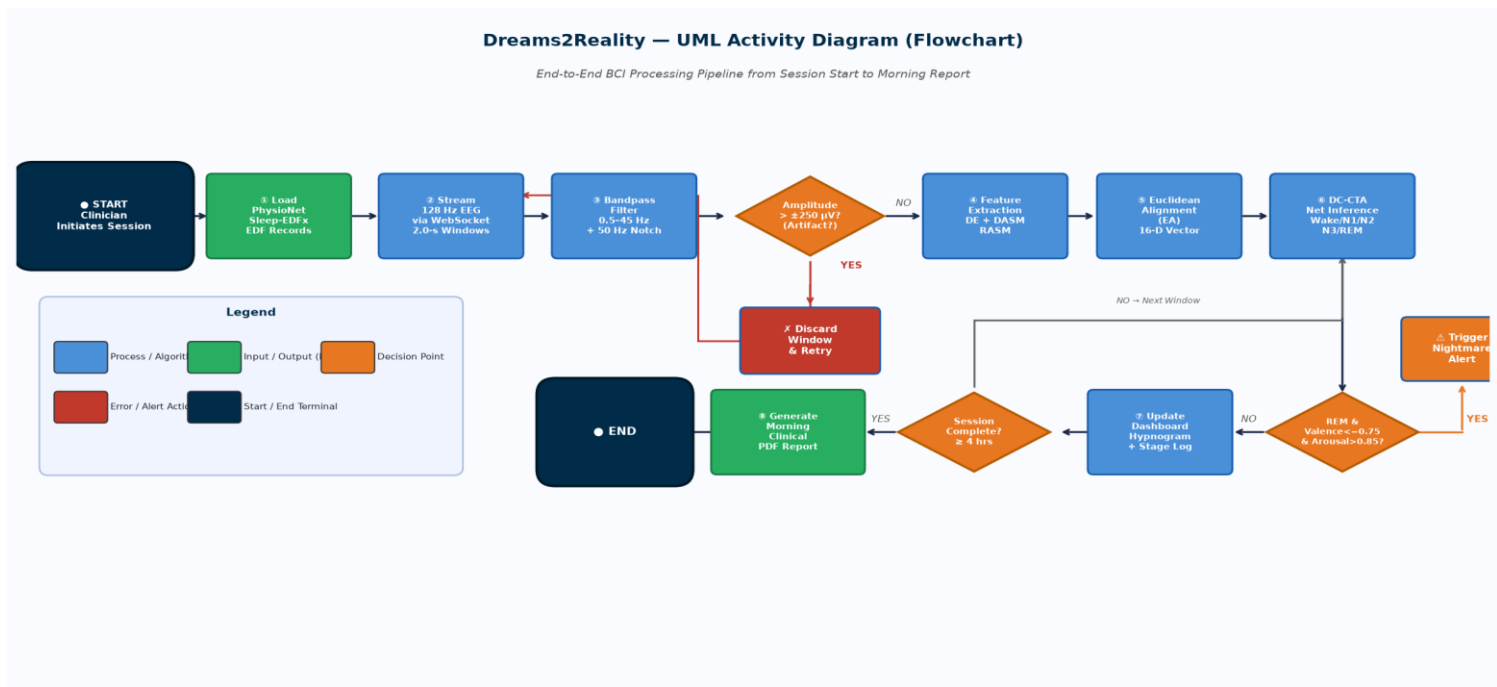


Figure: Dreams2Reality Activity & Swimlane Diagram — Four concurrent execution partitions

2.3 Data Flow Diagrams (DFDs)

2.3.1 DFD Level 0 & Level 1 — Context and Modular Pipeline

The **DFD Level 0 Context Diagram** establishes the global system boundary of Dreams2Reality, identifying external data producers (Sleep-EDFx Playback Streamer) and human consumers (Clinical Neurologist). The **DFD Level 1** decomposes the platform into five modular processes (Ingestion, Pre-processing, Feature Engineering, Inference, Reporting) interacting with four primary data stores.

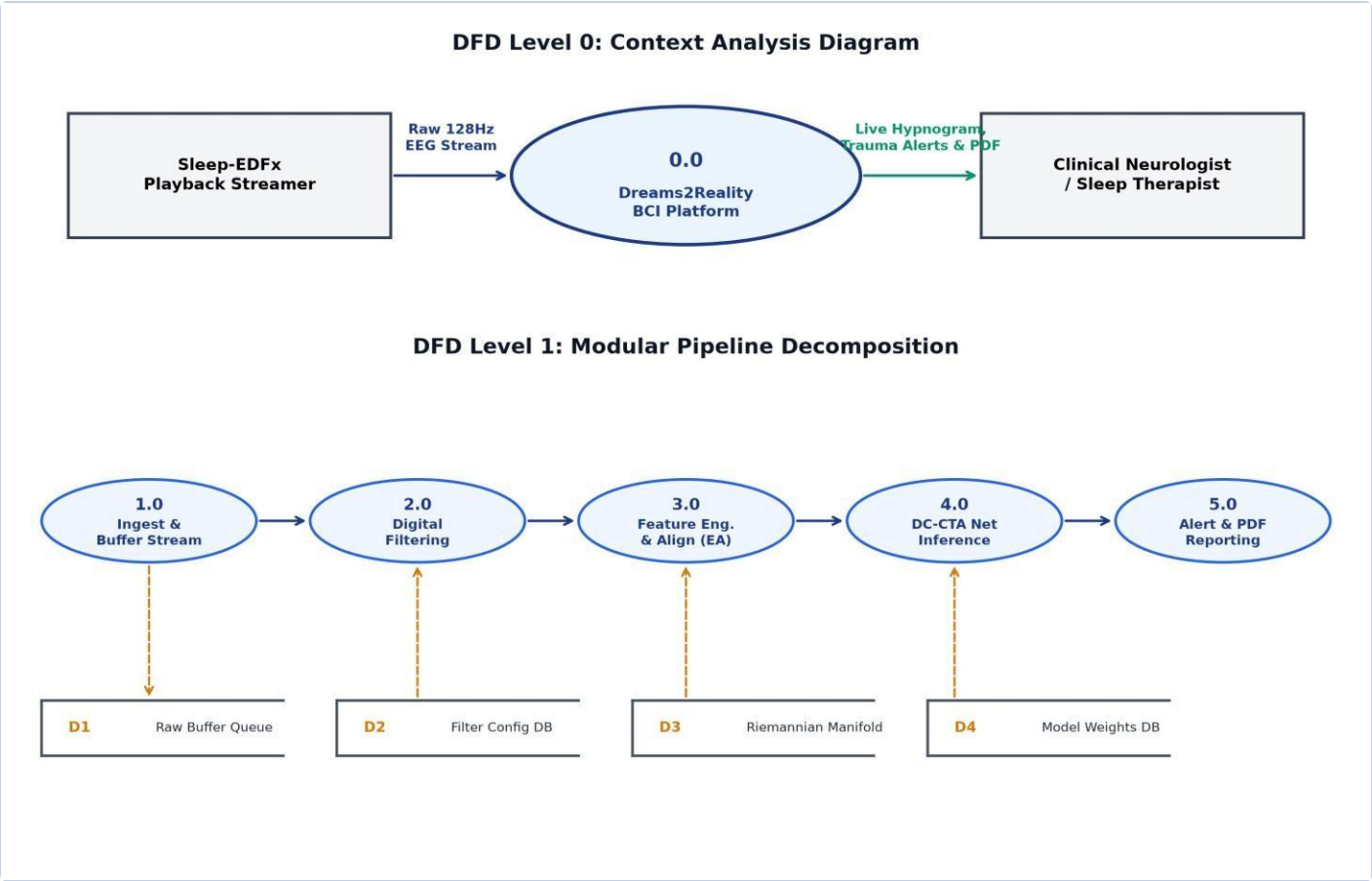


Figure: DFD Level 0 (Context Diagram) and DFD Level 1 (Modular Pipeline Decomposition) — combined view

2.3.2 DFD Level 2 — Feature Engineering & Spatial Alignment Subsystem (Process 3.0)

Level 2 provides a detailed decomposition of Process 3.0, showing how each 2.0-second EEG epoch is transformed from raw filtered samples into a 16-dimensional normalized neural feature tensor through six sequential sub-processes.

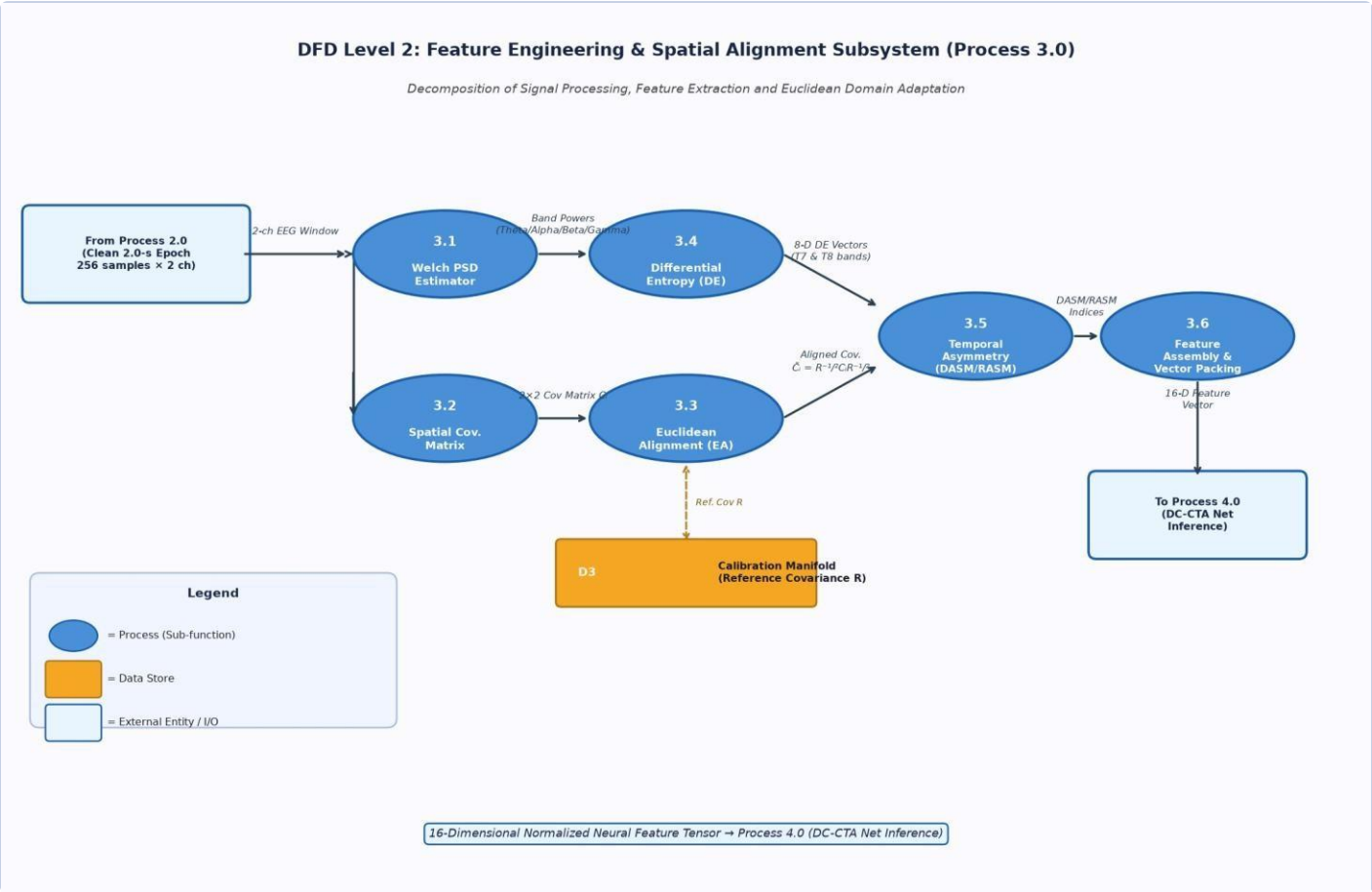


Figure: DFD Level 2 — Feature Engineering & Euclidean Alignment Subsystem, sub-processes 3.1 through 3.6

2.4 UML Class Diagram

The UML Class Diagram models the object-oriented software architecture of Dreams2Reality, identifying the core classes, their attributes, methods, and inter-class relationships.

UML Class Diagram: Dreams2Reality Object-Oriented Architecture

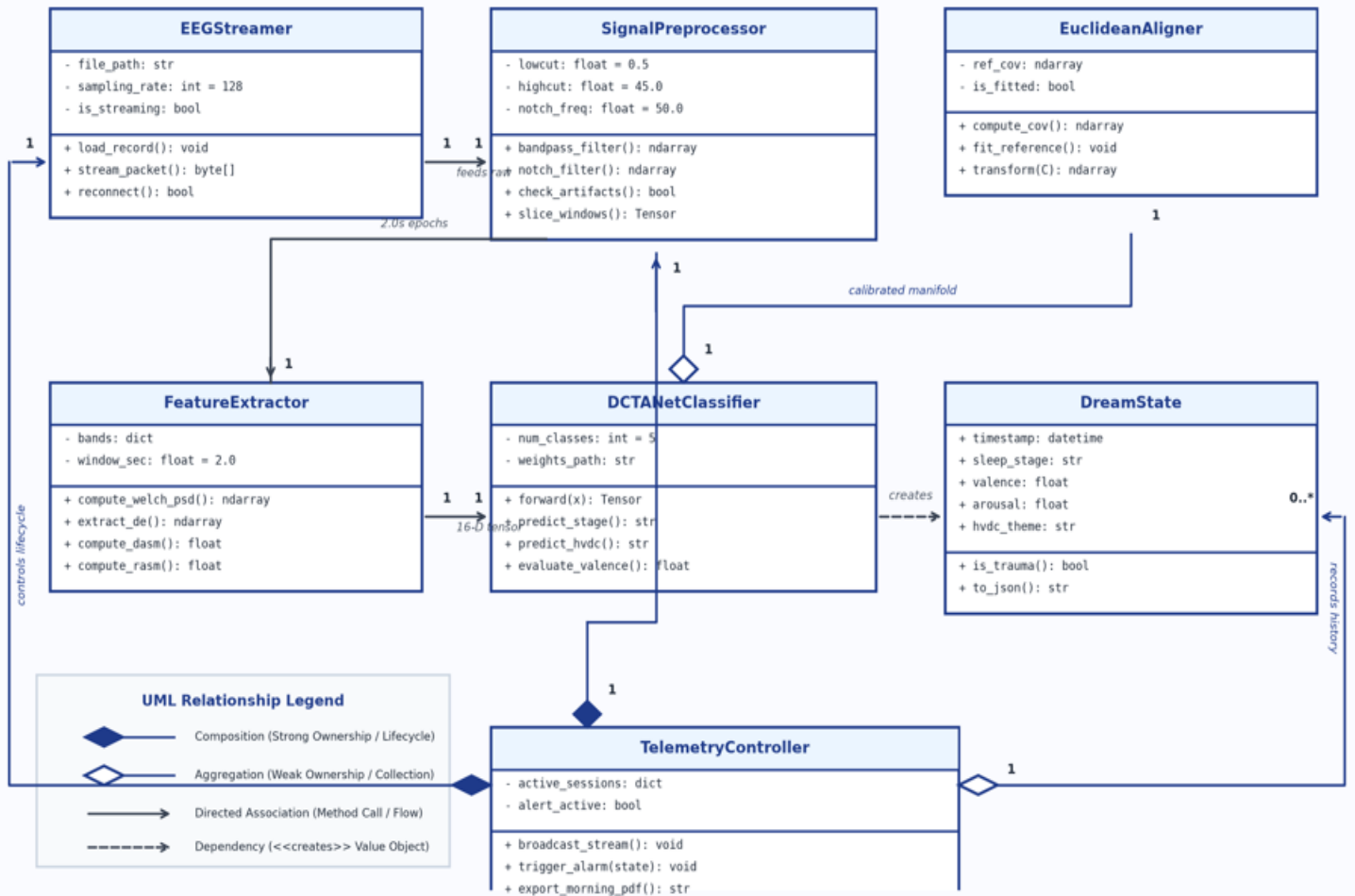


Figure: Dreams2Reality UML Object-Oriented Class Diagram — Core classes, attributes, methods, and relationships

2.5 Software Requirement Specification (SRS) in IEEE Format

1. Introduction

1.1 Purpose

This document specifies the software requirements for **Dreams2Reality**, a dual-channel Brain-Computer Interface (BCI) system designed for real-time sleep-stage classification, dream-state affective decoding, and bedside trauma monitoring. This specification conforms to the **IEEE Std 830-1998** standard for Software Requirements Specifications.

1.2 Scope

Dreams2Reality is a medical-grade assistive software application. The software ingests simulated or live two-channel digital EEG signals (T7–T8) sampled at 128 Hz, executes zero-phase bandpass filtering, extracts spectral and non-linear differential entropy features, projects spatial covariance matrices onto a calibrated Riemannian reference space, and classifies sleep/dream states using an attention-based neural network. The system delivers real-time telemetric visualization, generates acoustic/visual nightmare distress alerts, and compiles post-sleep clinical summary reports. Physical electrode hardware manufacturing is outside the scope of this project; telemetric input is simulated via benchmark clinical datasets.

1.3 Definitions, Acronyms, and Abbreviations

- **BCI:** Brain-Computer Interface
- **EEG:** Electroencephalogram
- **PSG:** Polysomnography
- **REM:** Rapid Eye Movement sleep
- **NREM:** Non-Rapid Eye Movement sleep (stages N1, N2, N3)
- **DE:** Differential Entropy
- **DASM:** Differential Asymmetry ($DE_{left} - DE_{right}$)
- **RASM:** Rational Asymmetry (DE_{left} / DE_{right})
- **EA:** Euclidean Alignment (Covariance matrix standardization)
- **HVdC:** Hall-Van de Castle standard dream content coding system
- **UAT:** User Acceptance Testing
- **FPS:** Frames Per Second (telemetric refresh rate)

1.4 References

- IEEE Std 830-1998: *IEEE Recommended Practice for Software Requirements Specifications*.
- Kemp, B. et al. (2000): *Analysis of a sleep-dependent neuronal feedback loop: the Sleep-EDF database*. IEEE Transactions on Biomedical Engineering.
- Davidson, R. J. (2004): *What does the prefrontal cortex do in affect: perspectives on frontal EEG asymmetry*. Biological Psychology.
- Wolpaw, J. R. et al. (2002): *Brain-computer interfaces for communication and control*. Clinical Neurophysiology.
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2. Overall Description

2.1 Product Perspective

Dreams2Reality is a self-contained, client-server biomedical monitoring package. The server component, built on Python and FastAPI, manages real-time mathematical signal transformations and neural inference. The client component, built on modern web standards (React/HTML5 Canvas), provides a low-latency graphical user interface for sleep laboratories and bedside hospital wards.

2.2 User Characteristics

Sleep Technicians & Neurologists: Possess professional clinical expertise in interpreting electroencephalographic rhythms and hypnograms. Require precise numerical metrics, raw wave views, and clear alert configurations.

Bedside Attendants & Caregivers: Non-specialist operators requiring intuitive, color-coded visual indicators of patient distress without complex technical jargon.

2.3 General Constraints

Compute Constraints: Must execute in real time on standard consumer hardware (minimum 4-core x86_64 or ARM64 processor, 4 GB RAM) without requiring dedicated discrete GPUs.

- **Latency Constraints:** End-to-end processing latency from window closure to UI dashboard alert must not exceed 15 ms.
 - **Sampling Rate:** Software pipeline assumes a fixed nominal sampling frequency of 128 Hz.
-

3. Specific Requirements

3.1 Functional Requirements

FR-01: Asynchronous Signal Telemetry Ingestion

The system shall ingest dual-channel electroencephalographic time-series data at 128 Hz over an asynchronous WebSocket connection from an automated file-based playback daemon or simulated hardware client.

FR-02: Zero-Phase Digital Filtering

The system shall apply a 4th-order zero-phase Butterworth bandpass filter passing frequencies between 0.5 Hz and 45.0 Hz, complemented by a 50.0 Hz infinite impulse response (IIR) notch filter to suppress electrical interference.

FR-03: Dynamic Artifact Rejection

The system shall continuously calculate peak-to-peak amplitude within each 2.0-second window and automatically flag/discard windows exceeding $\pm 250 \mu\text{V}$ as gross physiological motion artifacts.

FR-04: Sliding Window Assembly

The pre-processor shall segment continuous telemetry into 2.0-second epochs (256 samples per channel) with a configurable step stride of 0.5 seconds (75% overlap) to support low-latency sliding evaluation.

FR-05: Non-Linear Feature Computation

For each window, the system shall compute Welch Power Spectral Density and Differential Entropy across Theta (4–8 Hz), Alpha (8–12 Hz), Beta (12–30 Hz), and Gamma (30–45 Hz) frequency sub-bands.

FR-06: Hemispheric Asymmetry Indices

The system shall compute Differential Asymmetry (DASM) and Rational Asymmetry (RASM) metrics comparing homologous electrodes T7 and T8 to serve as affective valence descriptors.

FR-07: Euclidean Space Covariance Alignment

The system shall normalize the spatial covariance matrix of incoming epochs using a reference Riemannian mean covariance matrix to eliminate inter-subject cranial impedance drift.

FR-08: Sleep-Stage & Dream Classification

The neural inference engine shall output posterior probabilities for five clinical sleep states (Wake, N1, N2, N3, REM) and categorize REM dream mentation into Hall-Van de Castle archetypes (Motor Agitation, Threat/Panic, Vivid Sensory, Cognitive, Restful Baseline).

FR-09: Automated Trauma / Nightmare Alarm Generation

The system shall trigger a high-priority bedside auditory and visual alert whenever affective valence remains below -0.75 and arousal exceeds 0.85 during confirmed REM sleep for > 10 consecutive seconds.

FR-10: Morning Clinical Summary Export

The system shall generate an automated post-sleep clinical report in PDF format containing the macrostructural hypnogram, total dream duration, trauma episode timestamps, and natural language descriptions.

3.2 Non-Functional Requirements

NFR-01: Performance & Inference Latency

The end-to-end software inference latency-encompassing digital filtering, feature extraction, Euclidean projection, and forward neural pass-shall not exceed 15 milliseconds per 2.0-second epoch on an Intel Core i5 / Apple M1 or equivalent edge processor.

NFR-02: Pipeline Throughput

The system shall sustain a minimum processing throughput of 250 epochs per second during batch historical file processing, facilitating rapid retrospective overnight dataset scoring.

NFR-03: System Reliability & Availability

The core telemetry server daemon shall maintain an uptime availability of 99.5% over continuous 10-hour nocturnal monitoring sessions without memory leaks or buffer exhaustion.

NFR-04: Data Privacy & Security

All persisted patient telemetry, hypnogram records, and diagnostic dream summaries shall be encrypted at rest using AES-256 and transmitted locally over authenticated TLS/WSS channels.

NFR-05: User Interface Usability

The bedside dashboard interface shall refresh waveform visualizers at ≥ 30 FPS without frame stutter and shall display distress alerts in high-contrast red/amber typography compliant with WCAG 2.1 AA accessibility standards.

NFR-06: Software Maintainability & Modularity

The codebase shall be structured into decoupled object-oriented modules with a minimum of 85%unit test coverage across signal processing and alignment algorithms.

2.6 User Stories and Story Cards

The software engineering methodology utilizes an Agile/Scrum framework with two-week sprint cycles. Below are the five primary user stories guiding the Analysis and Prototyping phases.

Story Card 1: Real-Time Telemetry Streaming & Ingestion

- - **Story ID:** US-01 **User**
Story:

As a Sleep Technologist,

I want to stream 2-channel temporal EEG data at 128 Hz from simulated playback files, So that I can evaluate patients without needing physical 64-channel wet electrode caps.

Acceptance Criteria:

1. The system establishes a WebSocket connection within 500 ms of session launch.
2. Data packets arrive at exactly 128 samples/second with zero packet drop over 1 hour.
3. Dropped socket connections trigger an automated reconnect attempt within 3 seconds.
 -
 - **Priority:** Highest (Sprint 1)

Story Card 2: Dual-Channel Digital Signal Preprocessing

- - **Story ID:** US-02 **User**
Story:

As a Biomedical Data Scientist,

I want raw EEG signals to be automatically filtered between 0.5 Hz and 45 Hz and notch-filtered at 50 Hz, *So that* environmental powerline hum and baseline respiration drifts are purged before inference.

Acceptance Criteria:

1. The filter demonstrates zero phase distortion using forward-backward IIR/Butterworth implementation.
 2. Motion artifacts with amplitude $> \pm 250 \mu\text{V}$ are flagged and excluded from downstream inference.
 3. Pre-processing latency per 2.0-second epoch is under 3.0 ms.
 - **Priority:** High (Sprint 1)
 - **Estimation:** 3 Story Points
-

Story Card 3: Cross-Subject Euclidean Covariance Alignment

- - **Story ID:** US-03 **User**
Story:

As a Software Engineer,

I want to project temporal covariance matrices into a standardized Euclidean space,
So that the pre-trained neural network performs accurately on new patients without manual retraining.

Acceptance Criteria:

1. Computes the geometric Riemannian mean covariance matrix from a 60-second baseline epoch.
 2. Normalizes incoming covariance matrices via inverse matrix square root transformations.
 3. Maintains cross-subject zero-shot classification accuracy above 70% across diverse dataset subjects.
 - **Priority:** High (Sprint 2)
 - **Estimation:** 8 Story Points
-

Story Card 4: Automated Nightmare Distress Alert System

-
- **Story ID:** US-04 **User**
Story:

As a Bedside Caregiver,

I want to receive instant visual and acoustic warnings when a patient experiences intense nightmare distress, *So that* I can provide timely bedside comfort or therapeutic intervention.

Acceptance Criteria:

1. Triggers alert within 1.0 second of detecting REM negative valence (< -0.75) and high arousal (> 0.85).
2. The bedside UI displays a flashing high-contrast amber banner and emits a gentle auditory notification.
3. Attendant can mute or dismiss the alarm with a single touch/click, logging acknowledgment time

Story Card 5: Morning Clinical Dream Summary Report

-
- **Story ID:** US-05 **User**
Story:

As a Clinical Neurologist,

I want an automated PDF report summarizing the night's sleep stages and dream categories, *So that* I can evaluate sleep architecture and correlate trauma episodes without manual hypnogram scoring.

Acceptance Criteria:

1. Generates a multi-page PDF within 5 seconds of session conclusion.
 2. Report includes a visual hypnogram graph, sleep stage duration table, and nightmare timestamps.
 3. Categorizes dream mentation into Hall-Van de Castle semantic themes with clear percentage breakdowns.
- **Priority:** Medium (Sprint 3)
 - **Estimation:** 5 Story Points
-